

Computer vision

Image enhancement/
Image processing

Image Enhancement

Image Enhancement

The principal objective of enhancement is to process an image so that **the result is more suitable** than the original image for a **specific application**.

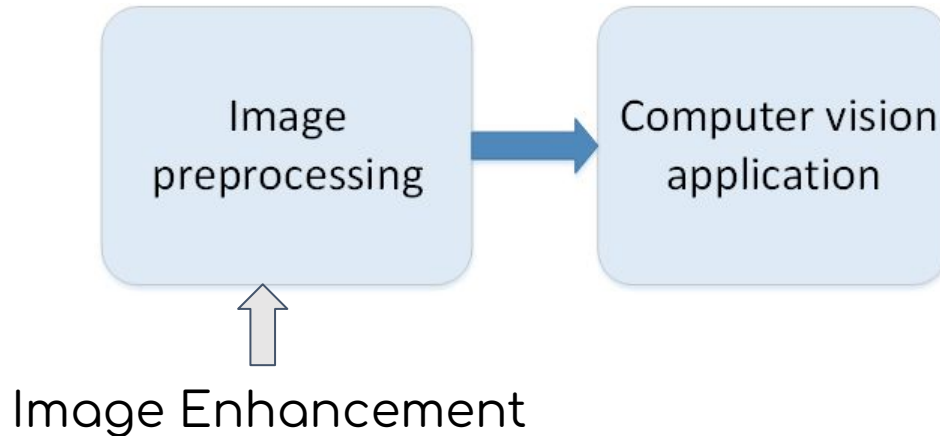


Image Histogram & Intensity Transformation

Unnormalized/normalized histogram

Let r_k , for $k = 0, 1, 2, \dots, L-1$, denote the intensities of an L-level digital image, $f(x, y)$.

The **unnormalized histogram** of f is defined as

$$h(r_k) = n_k \text{ for } k = 0, 1, 2, \dots, L-1$$

where n_k is the number of pixels in f with intensity r_k , and the subdivisions of the intensity scale are called histogram bins.

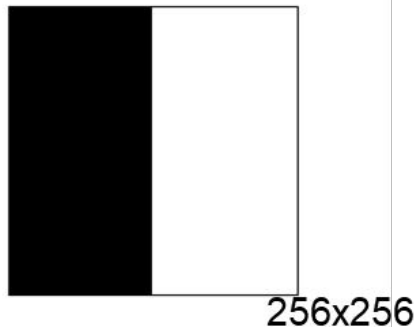
The **normalized histogram** of f is defined as

$$\rho(r_k) = h(r_k) / (M.N)$$

where, M and N are the number of image rows and columns, respectively.

Histogram example

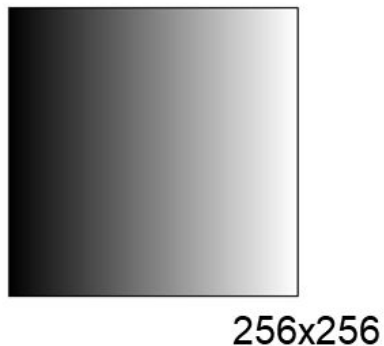
bi-level image



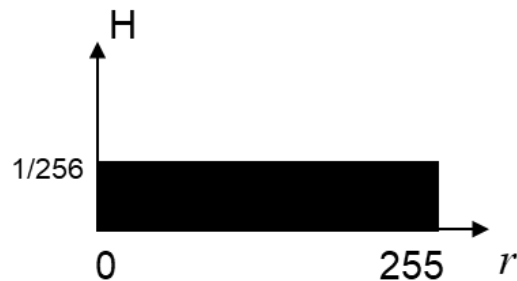
histogram



Pixel values linearly increasing from 0 to 255 with increasing column index

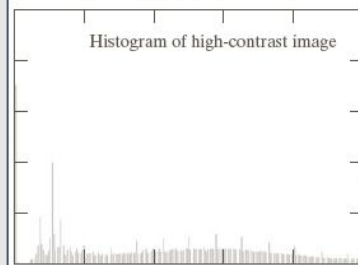
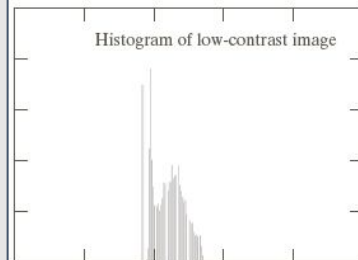
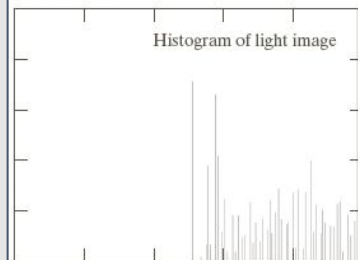
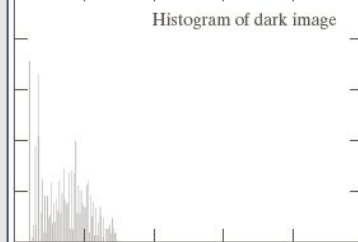
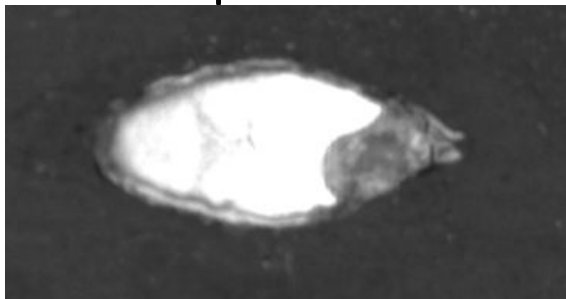


histogram



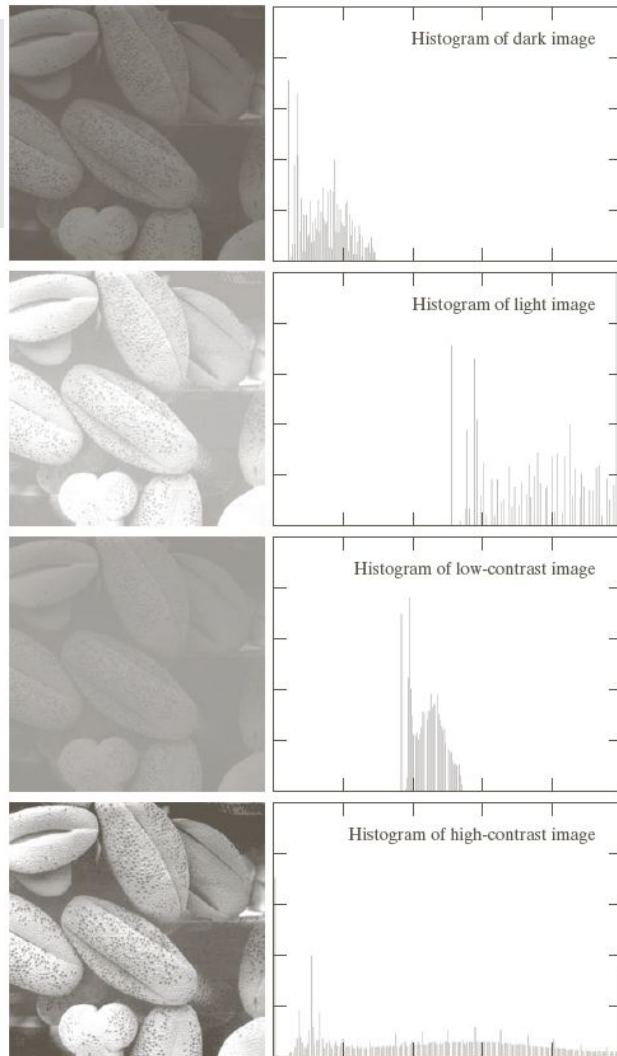
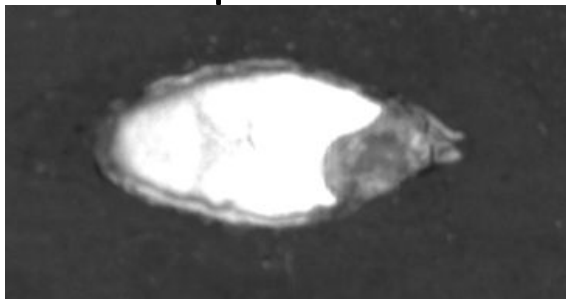
Brightness & Contrast in histogram

Examples:



Brightness & Contrast in histogram

Examples:



Intensity Transformation

Intensity Transformation

Intensity transformation is a basic digital image processing technique, where the pixel intensity levels of an image are transformed to new values using a mathematical transformation function.

$$s=T(r)$$

where:

s is the new pixel intensity value

r is the original pixel intensity value , $r \geq 0$

T(r) is a transformation function

Common Methods of Image Intensity Transformation

- For intensity transformation of an image, various **mathematical operations** can be used, such as
 - negation $s = \text{MaxVal} - r$
 - multiplication $s = r * c$
 - constant addition $s = r + c$
 - logarithmic function $s = \log(\text{const} + r)$
 - power-law (gamma) transformation $s = c * r^\gamma$
 - and more...

where s is the new pixel intensity value, r is the original pixel intensity value, c is constant (parameter)

- Intensity transformation using **Histogram Equalization**

Pixels intensity direct manipulation

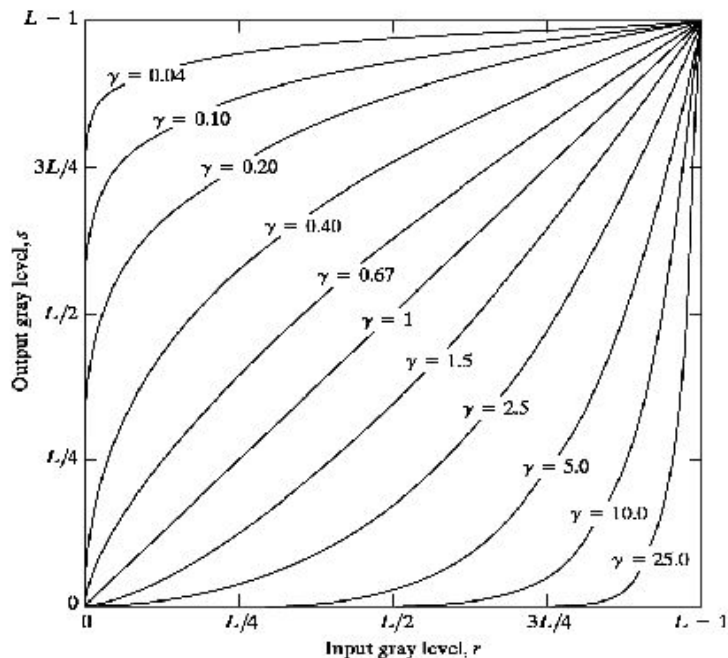
Power-Law Transformations

Power-law transformations have the basic form:

$$s = cr^\gamma$$

where c and γ are positive constants.

Power-Law Transformations



Plots of the equation

$$s = cr^\gamma$$

various values of γ

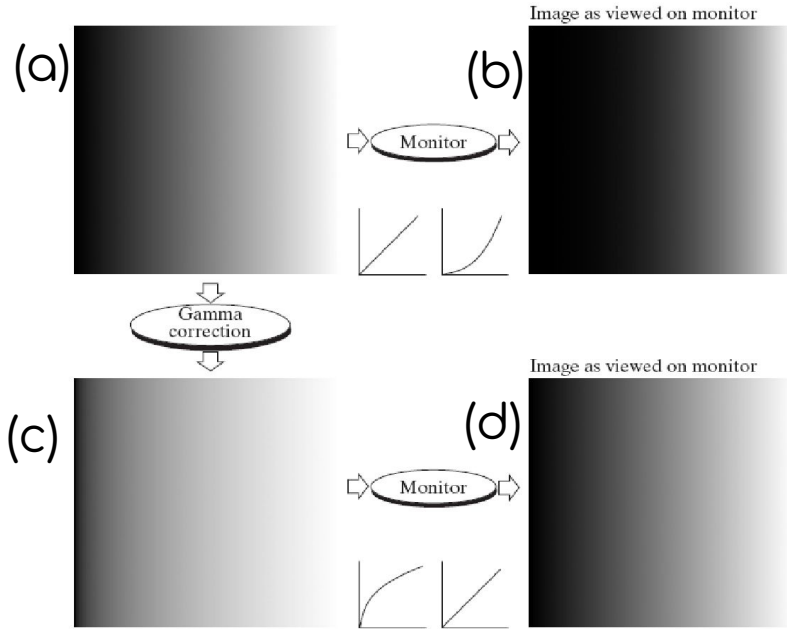
Signal range: 0-1

Power-Law Transformations used for Gamma Correction

The process of using the Power-Law Transformations is called gamma correction.

For example, cathode ray tube (CRT) devices have an intensity-to-voltage response that is a power function, with exponents varying from approximately 1.8 to 2.5

Tube Monitor Gamma Correction



(a) Linear scale gray-scale image input.

(b) Output of monitor

(c) Gamma corrected gray-scale input

(d) Output of monitor.

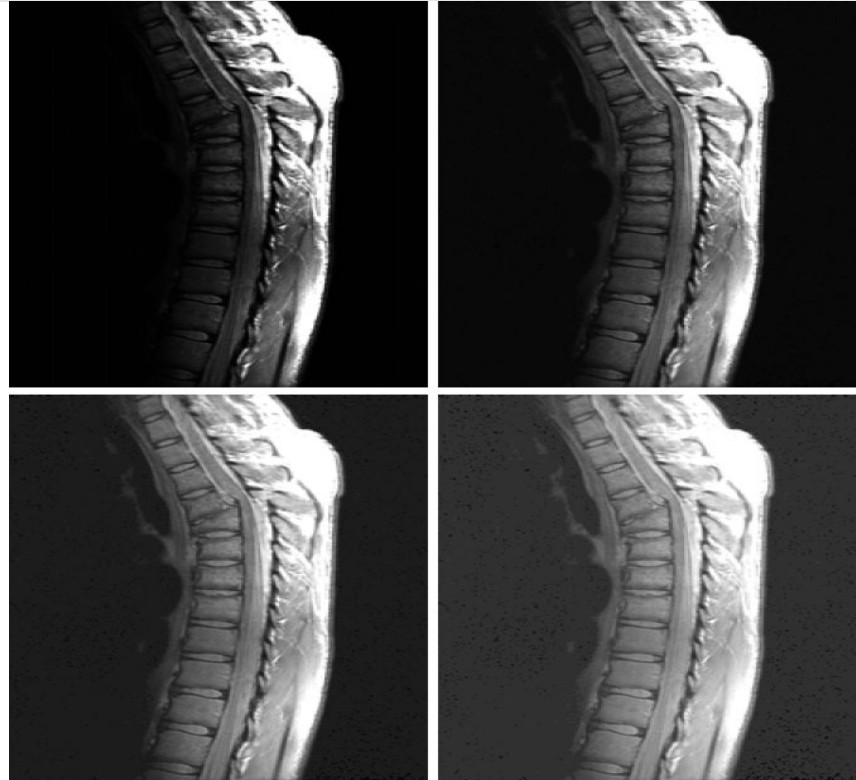
Tube monitor gamma correction adjusts the non-linear relationship between a CRT's input voltage and light output.

This is typically a power function with an exponent (gamma) around 2.5.

Power-Law Transformations - example

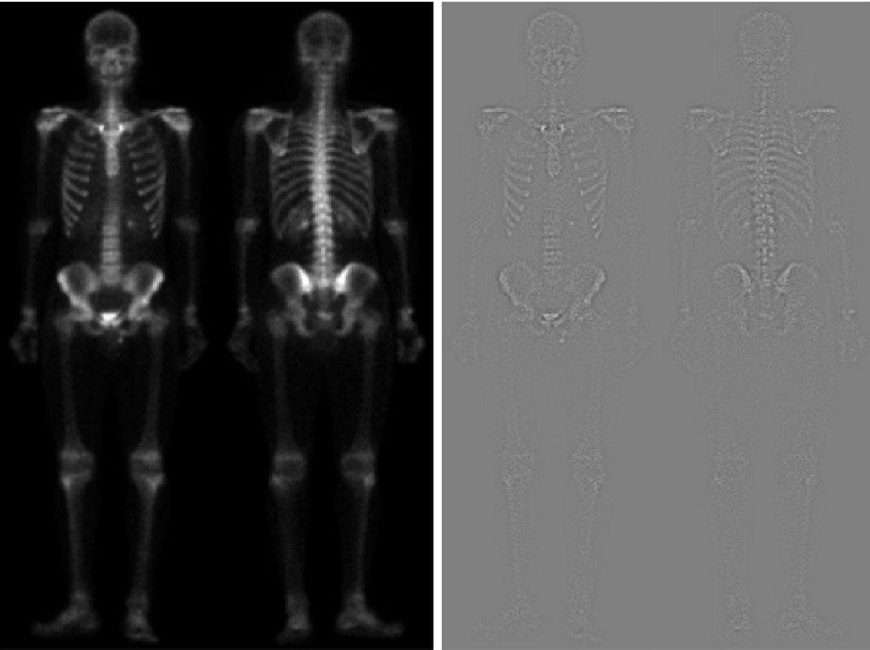
(a) Magnetic resonance (MR) image of a fractured human spine.

(b)–(d) Results of applying the power-law transformation with $c=1$ and $\gamma=0.6, 0.4$, and 0.3 , respectively.



Combining Spatial Enhancement Methods

Example of Combining Spatial Enhancement Methods



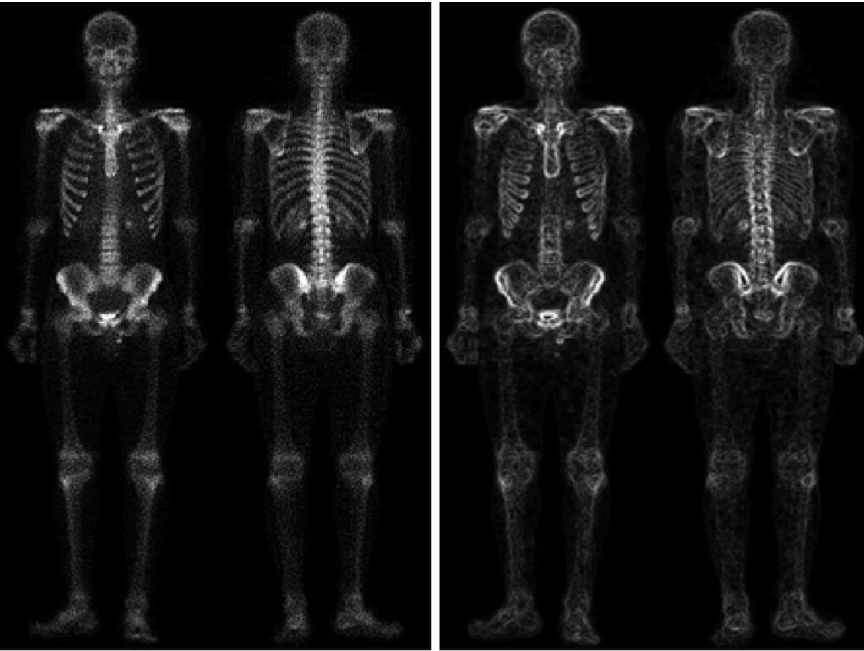
(a) Image of whole body bone scan.

(b) Laplacian of (a)

(a)

(b)

Example of Combining Spatial Enhancement Methods



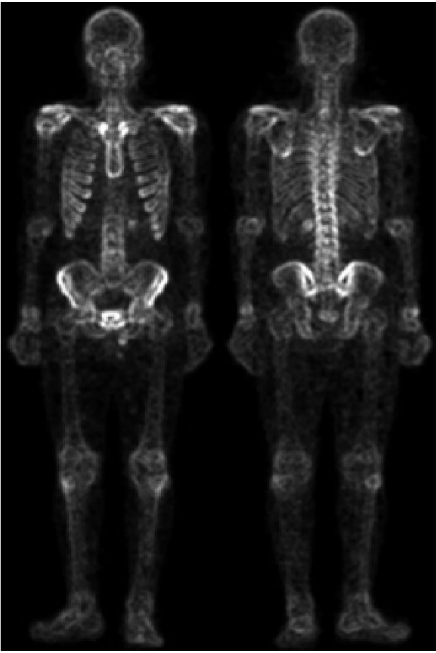
(c)

(d)

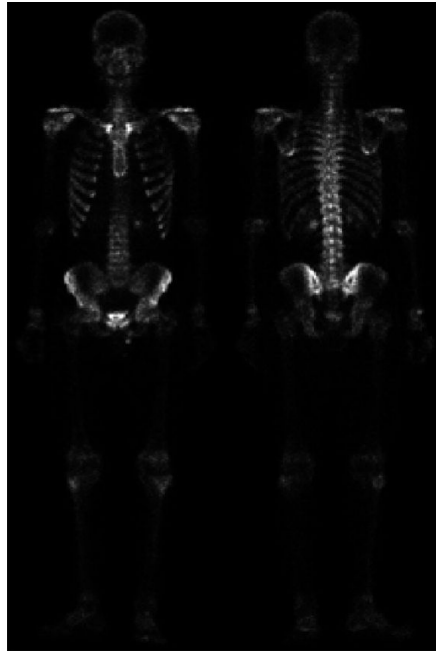
(c) Sharpened image obtained by adding (a) and (b).

(d) Sobel of (a)

Example of Combining Spatial Enhancement Methods



(e)

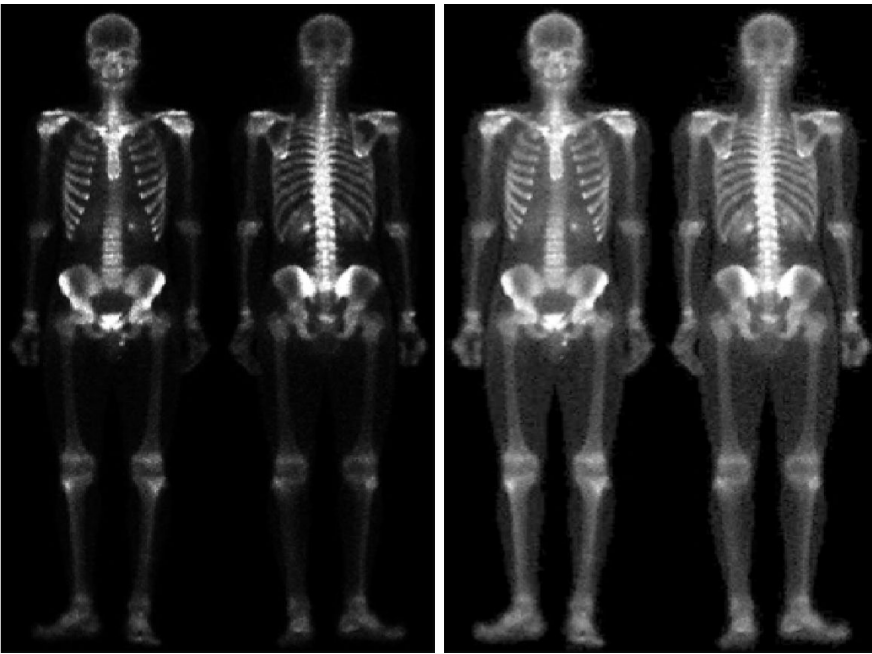


(f)

(e) Sobel image smoothed with a 5×5 averaging filter.

(f) Mask image formed by the product of (c) and (e).

Example of Combining Spatial Enhancement Methods



(g)

(h)

(g) Sharpened image obtained by the sum of (a) and (f).

(h) Final result obtained by applying a power-law transformation to (g).

Histogram Equalization

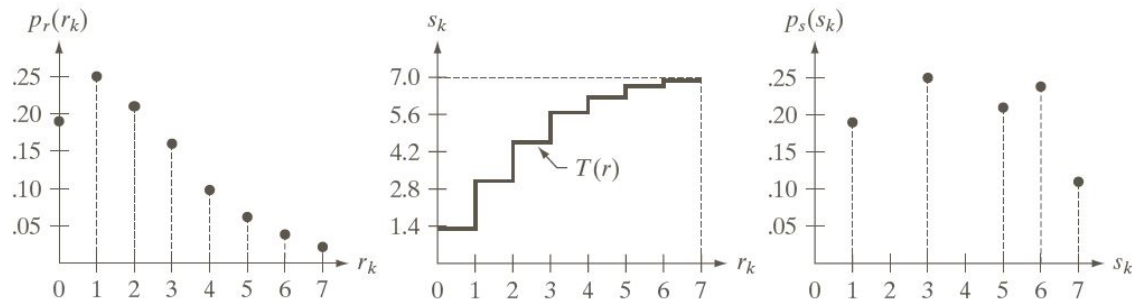
Histogram Equalization - global

Histogram equalization involves transforming the intensity values so that the histogram of the output image approximately matches a specified histogram.

$T(r)$ is transformation function.

$T(r)$ could be calculated using the Cumulative Distributive function of the image

Example:



a b c

FIGURE 3.19 Illustration of histogram equalization of a 3-bit (8 intensity levels) image. (a) Original histogram. (b) Transformation function. (c) Equalized histogram.

Resulting histogram when trying to achieve uniform* equalized histogram

*Resulting histogram is usually not uniform due to discrete nature of transform.

Contrast Limited Adaptive Histogram Equalization (CLAHE)

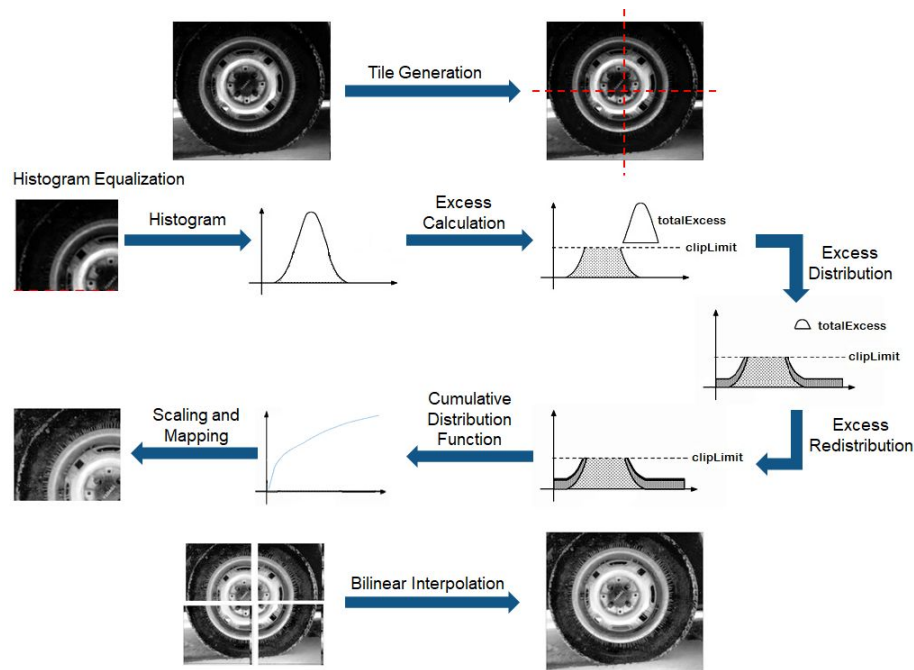
1. Dividing the image into blocks
2. Computing the histogram for each block
3. Histogram clipping:

If any histogram bin exceeds a predefined **maximum threshold**, it is clipped to prevent excessive contrast amplification. The clipped excess is then redistributed evenly across all histogram bins, reducing noise in uniform areas.

4. Local histogram equalization:

After adjusting the histogram, histogram equalization is applied to each block. This is done using the cumulative distribution function (CDF), which redistributes pixel intensity levels more uniformly.

5. Bilinear interpolation between blocks
6. Merging the processed blocks



CLAHE advantages/disadvantages

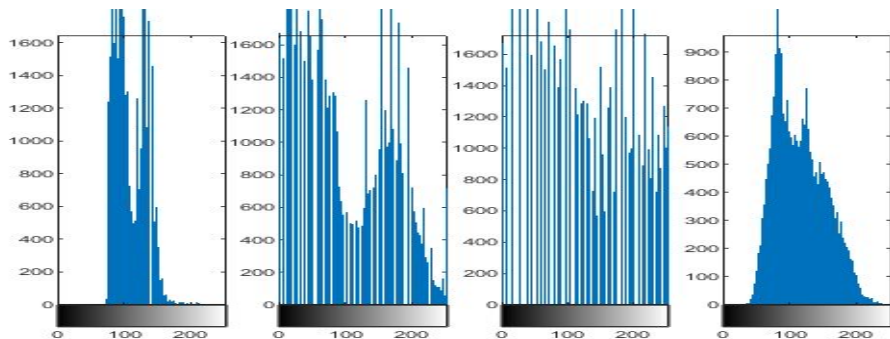
Advantages of the CLAHE transform are the modest computational requirements, its ease of use (requiring only one parameter: the clip limit), and its excellent results on most images.

Disadvantages of CLAHE:

- there is no 1 to 1 relationship between the gray values of the original image and the CLAHE processed result; -> CLAHE images are not suited for quantitative measurements that rely on a physical meaning of image intensity.
- artifacts that sometimes occur when high-intensity gradients are present

Image contrast enhancement -example

Original Image and Enhanced Images using imadjust, histeq, and adapthisteq



1. original
2. adjust the contrast of the image, specifying contrast limits
3. enhance contrast using histogram equalization
4. contrast-limited adaptive histogram equalization (CLAHE)